Probability and Statistics 1

Tutorial Sheet — Session 2

Frequency distribution of a quantitative character; classes, histogram, polygon and cumulative curve

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Exercise 1

The number of GPUs per node was observed on a sample of $n = 60$ nodes of a compute cluster. The results are:

GPUs x_i	1	2	4	8	16
n_i	12	20	16	9	3

(a) What is the type of the character? Justify.

(b) Complete the frequency table with the relative frequencies f_i, the percentages p_i, the increasing cumulative absolute frequencies $n_i^{\nearrow}$ and the increasing cumulative relative frequencies $f_i^{\nearrow}$.

GPUs x_i	n_i	f_i	p_i	$n_i^{\nearrow}$	$f_i^{\nearrow}$
1	12				
2	20				
4	16				
8	9				
16	3				
Total	60			–	–

(c) What is the mode of the distribution? Justify.

(d) How many nodes have at most 4 GPUs? At most 8 GPUs?

Exercise 2

Using the data of Exercise 1:

(a) Draw a bar chart of the distribution of the number of GPUs per node. State clearly what the horizontal and vertical axes represent, and where the vertical axis starts.

(b) Draw the polygon of frequencies of the distribution. What does the shape of the polygon tell you about the distribution?

(c) Why is a pie chart not appropriate for this character, whereas it was for the character of Session 1?

Exercise 3

The inference latency of a model, in milliseconds, was measured on a sample of $n = 100$ requests. The smallest value is 30 ms and the largest is 80 ms.

(a) What is the type of the character? Justify.

(b) Compute the range $r = x_{\max} - x_{\min}$ of the series.

(c) Using the rule of thumb $K = 1 + \frac{10}{3}\log_{10} n$, compute the number of classes. Round to the nearest integer.

(d) Deduce the common amplitude $a = r/K$ of the classes, and give the list of the classes (with intervals of the form $[a_i, b_i[$).

(e) Give the centre of each class.

Exercise 4

The latency data of Exercise 3, grouped in the five classes $[30;40[$, $[40;50[$, $[50;60[$, $[60;70[$, $[70;80[$, give the following absolute frequencies:

$$n_1 = 8, \qquad n_2 = 22, \qquad n_3 = 41, \qquad n_4 = 21, \qquad n_5 = 8.$$

(a) Complete the frequency table below (class, centre x_i, n_i, f_i, p_i, $n_i^{\nearrow}$).

Class	Centre x_i	n_i	f_i	p_i	$n_i^{\nearrow}$
$[30;40[$		8			
$[40;50[$		22			
$[50;60[$		41			
$[60;70[$		21			
$[70;80[$		8			
Total	–	100			–

(b) Check that the relative frequencies sum to 1 and the percentages to 100.

(c) What is the modal class? Justify.

(d) How many requests were served in less than 60 ms? In less than 50 ms?

Exercise 5

Using the data of Exercise 4:

(a) Draw the histogram of the distribution. State clearly what the horizontal and vertical axes represent. Compute the total area of the histogram and check that it equals $a\,n$, where a is the common amplitude of the classes.

(b) Draw the cumulative frequency curve, plotting $n_i^{\nearrow}$ against the upper bound of each class. What does the curve allow you to read directly?

(c) Using the cumulative curve, estimate the fraction of requests served in less than 55 ms. (Read on the graph, then check with a linear interpolation inside the class $[50; 60[$.)

Exercise 6

Two services, A and B, record the latency of 100 requests each. Both distributions are grouped in the classes $[0; 20[$, $[20; 40[$, $[40; 60[$, $[60; 80[$, $[80; 100[$, with the following frequencies:

	$[0; 20[$	$[20; 40[$	$[40; 60[$	$[60; 80[$	$[80; 100[$
Service A	40	35	15	7	3
Service B	10	15	25	30	20

(a) Complete the two frequency tables (relative frequencies only).

(b) Compute the increasing cumulative relative frequencies for each service.

(c) Sketch the two histograms on the same figure (or side by side) and the two cumulative curves on the same figure.

(d) Which service has the lower median latency? Justify your answer using the cumulative curves, without computing the medians exactly.

(e) In one or two sentences, describe the shape of each distribution (concentrated on small values, symmetric, spread towards large values).

Exercise 7

A dataset of $n = 250$ file sizes, in kilobytes, has a minimum of 0.5 kB and a maximum of 510 kB.

(a) Compute the range and propose a number of classes using the rule of thumb. Deduce a rounded amplitude a (choose a "convenient" value, e.g. a multiple of 10, 20 or 50).

(b) Give the list of classes.

(c) Could the first class be $[0; 50[$ even though the minimum is 0.5? Justify.

(d) In one or two sentences, explain what is lost when the raw data are replaced by a table of classes, and what is gained.